Lecture 6: Dependent Sources and the Ideal Transformer

EE201 – Circuit Theory I

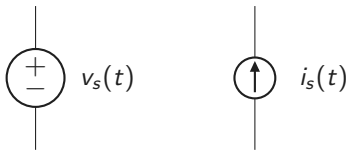
M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

current voltage power magnetic flux

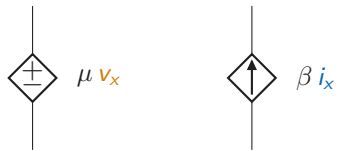
Sources controlled from elsewhere

independent (Lecture 3)



decides
its own value

dependent (today)



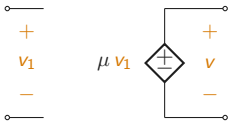
copies a variable
from elsewhere, scaled

Dependent (controlled) source

A voltage or current source whose value is a function of a voltage or current in *another* branch — the controlling variable. In EE201 the function is always a constant times that variable: a linear dependent source.

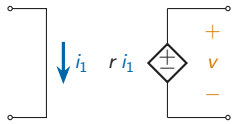
The four kinds

1. VCVS — voltage-controlled voltage source



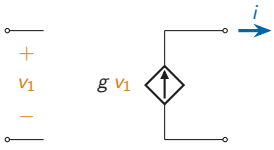
$v = \mu\, v_1$

2. CCVS — current-controlled voltage source



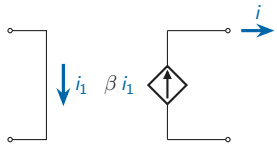
$v = r\, i_1$

3. VCCS — voltage-controlled current source



$i = g\, v_1$

4. CCCS — current-controlled current source



$i = \beta\, i_1$

The name reads in order: what it senses, then what it produces.

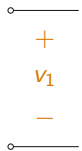
The constant, and its unit

Kind	Equation	The constant	Unit
VCVS	$v = \mu\, v_1$	voltage gain	none (V/V)
CCVS	$v = r\, i_1$	transresistance	ohm, Ω (V/A)
VCCS	$i = g\, v_1$	transconductance	siemens, S (A/V)
CCCS	$i = \beta\, i_1$	current gain	none (A/A)

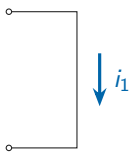
The unit is always what comes out over what goes in.

Trans- when the two differ: a transresistance turns a current into a voltage, so it is measured in ohms.

The sensing side



senses a voltage:
open circuit, $i = 0$

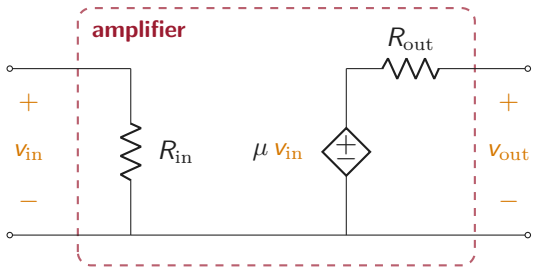


senses a current:
short circuit, $v = 0$

It senses without disturbing anything.

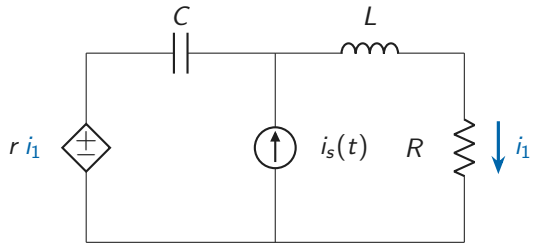
In real diagrams this side is not drawn: the controlling variable is the voltage across, or the current through, an element that is already there — labelled v_x, i_x, i_1.

Why we need them



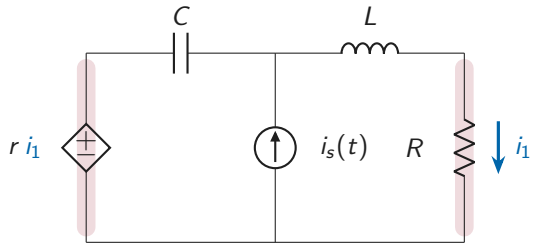
- The simplest model of anything that amplifies: the gain μ is the only place where the output can grow larger than the input.
- A **transistor** is modelled with a VCCS or a CCCS — in the electronics courses.
- An **op-amp** is a VCVS with $\mu \approx 10^5$ — later in this course.

Example: what the diamond says



Which equation does the dependent source contribute?

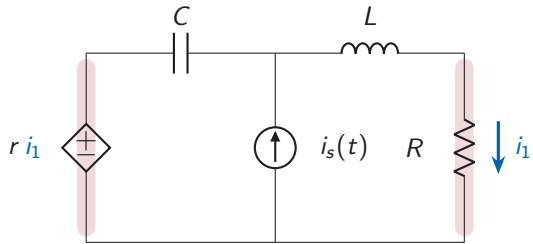
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Which equation does the dependent source contribute?

A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

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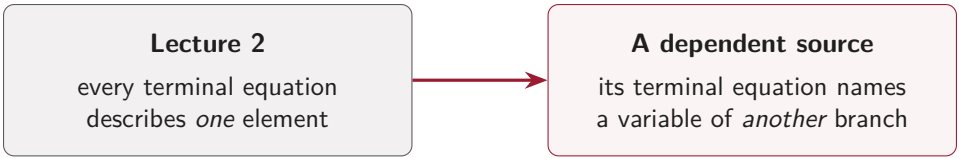


Which equation does the dependent source contribute?

A CCVS: $v = r\, i_1$ — and i_1 flows through R, at the far right.

$n = 4$, $b = 5$: still $2b = 10$ unknowns and $b = 5$ terminal equations.
Four speak about one element each. The fifth links two branches.

What is new — and what is not



KCL and KVL are unchanged. Still $2b$ equations for $2b$ unknowns.

Two meanings of "controlled"

Lecture 3: a *voltage-controlled resistor* has $i = f(v)$ of its own voltage. A voltage-controlled *source* takes its value from another branch.

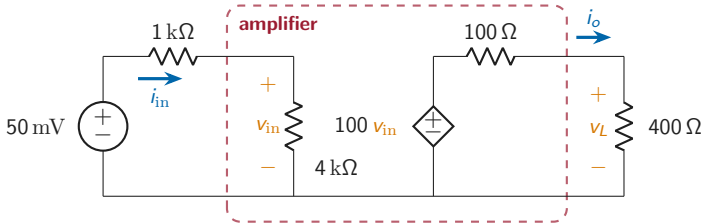
No new method

Recipe

1. Treat the dependent source as if it were independent, with its value written as μv_x, $r i_x$, $g v_x$ or βi_x.
2. Write KCL, KVL and the terminal equations as usual.
3. Add one equation that gives the controlling variable in terms of your unknowns.

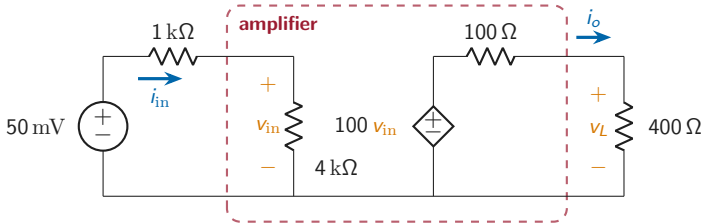
The power check $\sum_k p_k = 0$ still works — include the dependent source.

Example: an amplifier



A microphone drives an amplifier with $\mu = 100$. Find v_L.

Example: an amplifier



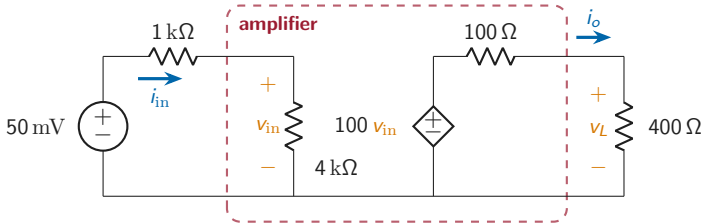
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Input side — a divider:

$$i_{in} = \frac{50\,\text{mV}}{5\,\text{k}\Omega} = 10\,\mu\text{A}$$

$$v_{in} = 4\,\text{k}\Omega \times 10\,\mu\text{A} = 40\,\text{mV}$$

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Output side — the diamond is $4\,\text{V}$:

$$i_o = \frac{4\,\text{V}}{500\,\Omega} = 8\,\text{mA}$$

$$v_L = 3.2\,\text{V}, \qquad \frac{v_L}{v_s} = 64$$

Where the power comes from

input side	
microphone	$-0.5\,\mu\mathrm{W}$
R_s	$+0.1\,\mu\mathrm{W}$
R_{in}	$+0.4\,\mu\mathrm{W}$

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output side	
dependent source	$-32\,\mathrm{mW}$
R_{out}	$+6.4\,\mathrm{mW}$
R_L	$+25.6\,\mathrm{mW}$

Each side sums to zero.

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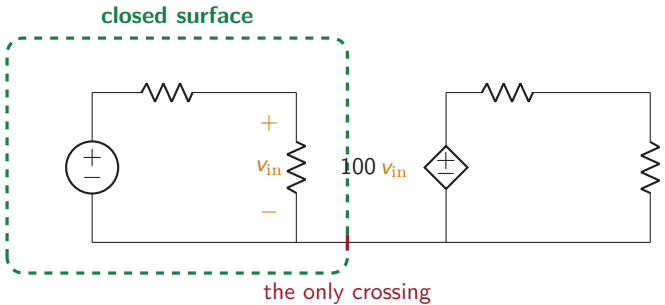
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The load gets 51 200 times what the microphone supplies.
The dependent source delivered it. A real amplifier takes it from its power supply, which the model does not draw.

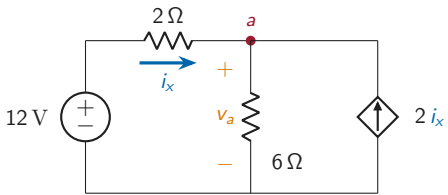
The shared wire carries no current



KCL on the surface (Lecture 2): one branch crosses it, so its current is zero.

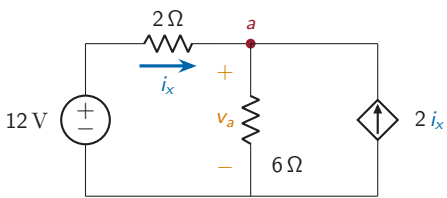
The two sides interact only through v_{in}.

Example: when the controlling variable is unknown



Find i_x and v_a.

Example: when the controlling variable is unknown



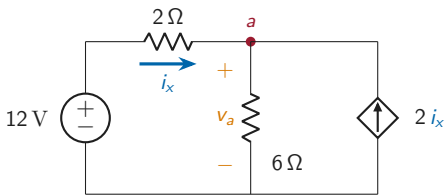
Find i_x and v_a.

Equations — i_x is already an unknown:

$$\text{KVL:}\quad -12 + 2i_x + v_a = 0$$

$$\text{KCL at } a\text{:}\quad -i_x - 2i_x + \tfrac{v_a}{6} = 0 \;\Rightarrow\; v_a = 18\, i_x$$

Example: when the controlling variable is unknown



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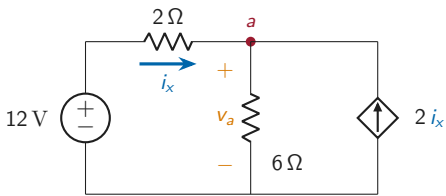
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Solution: $-12 + 20\, i_x = 0$

$$i_x = 0.6\,\text{A}, \qquad v_a = 10.8\,\text{V}$$

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Power check:

$$\underbrace{-7.2}_{12\,\text{V}} + \underbrace{0.72}_{2\Omega} + \underbrace{19.44}_{6\Omega} \underbrace{-\,12.96}_{\text{CCCS}} = 0$$

The CCCS carries $1.2\,\text{A}$ *out of* its + terminal: it delivers.

What a dependent source cannot do

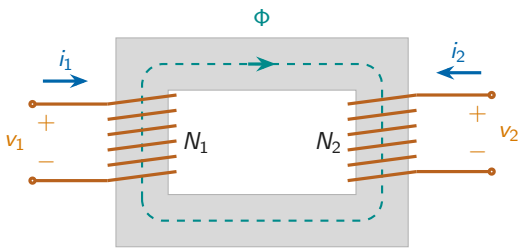
It cannot start anything

Switch the $12\,\mathrm{V}$ off: $2i_x + v_a = 0$ and $v_a = 18i_x$, so $20\,i_x = 0$.
Every voltage and current is zero — the dependent source's too.

It is never switched off

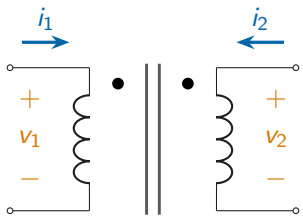
Later we switch sources off one at a time (superposition, Thévenin). Only independent sources are switched off: a dependent source is part of how the circuit behaves, not an input.

Two coils, one core



- **Lecture 1:** no changing magnetic field links the loops.
- **Lecture 2:** if we *want* such a link, draw it as an element.
- **Lecture 5:** a coil's field passes through the coil itself.
- **Today:** two coils on one iron core — the field of each passes through the other.

The ideal transformer



Two equations

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} \qquad N_1 i_1 + N_2 i_2 = 0$$

v positive and i entering at the dots.

Equivalently $v_2 = \frac{N_2}{N_1} v_1$, $i_2 = -\frac{N_1}{N_2} i_1$.

The dots act like reference marks. Move one dot and both equations change sign.

Where the two equations come from

Voltages: one flux through every turn

Each turn sees $d\Phi/dt$. N turns in series: $v = N\, d\Phi/dt$.

$$\frac{v_1}{N_1} = \frac{v_2}{N_2} = \frac{d\Phi}{dt}$$

Currents: an ideal core needs no current

The turn-weighted currents must cancel: $N_1 i_1 + N_2 i_2 = 0$.
A real core needs a small magnetising current; the ideal one needs none.

With k windings on one core: $\dfrac{v_1}{N_1} = \cdots = \dfrac{v_k}{N_k}$ and $N_1 i_1 + \cdots + N_k i_k = 0$.

No derivatives — and no DC

The equations are algebraic: no d/dt, no memory.

The ideal transformer is a resistive element — listed with resistors and dependent sources in the course outline, even though it is built from coils.

A real transformer needs a changing flux

Feed it DC and, once the current settles, the flux is constant: the secondary voltage is zero and the primary is just a coil — a short circuit at DC (Lecture 5). Use the ideal model for changing waveforms only.

The ideal transformer absorbs no power

Each winding obeys the PSC, so $v_j i_j$ is what it absorbs.

$$p(t) = \sum_{j=1}^{k} v_j i_j = 0 \qquad \text{at every instant}$$

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Call the common ratio $\lambda(t) = v_j / N_j$, so $v_j = N_j \lambda$:

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The first equation gave λ; the second made the sum vanish. Both are needed.

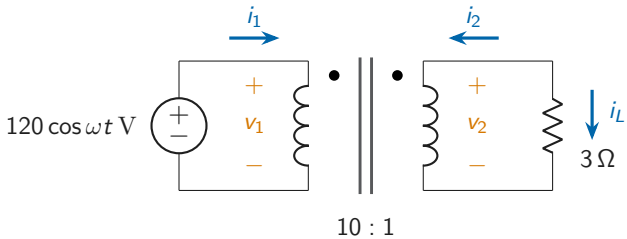
Dissipate, store, or pass through

Element	Absorbed power	The energy is
resistor	$p \geq 0$	dissipated
capacitor, inductor	p can change sign	stored, then returned
ideal transformer	$p = 0$ at every instant	passed straight through

Two windings: $v_1 i_1 = -v_2 i_2$.

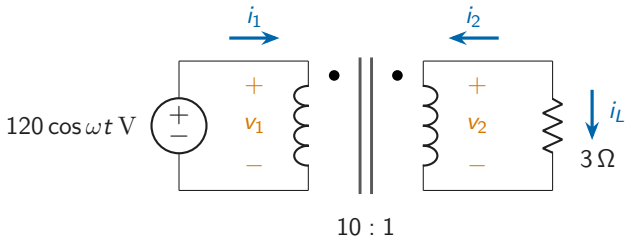
Voltage multiplied by N_2/N_1, current divided by N_2/N_1 — the product unchanged.

Example: a step-down transformer



Find the currents and the power on each side.

Example: a step-down transformer



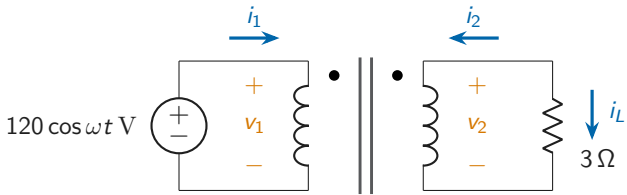
Find the currents and the power on each side.

$v_2 = \frac{1}{10} v_1 = 12\cos\omega t$ V

$i_L = v_2/3 = 4\cos\omega t$ A, so $i_2 = -4\cos\omega t$ A

$i_1 = -\frac{1}{10} i_2 = 0.4\cos\omega t$ A

Example: a step-down transformer



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$p_1 = v_1 i_1 = 48\cos^2\omega t$ W

$p_2 = v_2 i_2 = -48\cos^2\omega t$ W

$p_1 + p_2 = 0$ — at every instant.

Seen from the source

$$\frac{v_1}{i_1} = \frac{120}{0.4} = 300\,\Omega = 10^2 \times 3\,\Omega$$

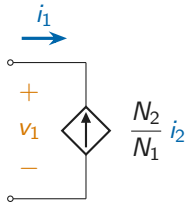
The source cannot tell the transformer and its $3\,\Omega$ load from a single $300\,\Omega$ resistor.

A transformer rescales resistance

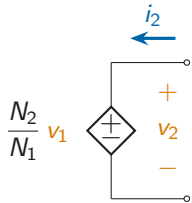
$$\frac{v_1}{i_1} = \left(\frac{N_1}{N_2}\right)^2 R_L$$

That is how an $8\,\Omega$ loudspeaker is made to look like a much larger resistance to the amplifier driving it.

A transformer is two dependent sources

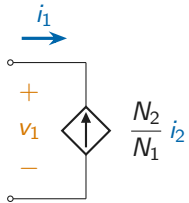


CCCS, controlled by i_2

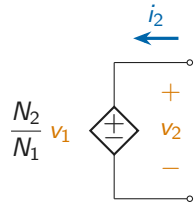


VCVS, controlled by v_1

A transformer is two dependent sources



CCCS, controlled by i_2

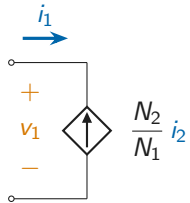


VCVS, controlled by v_1

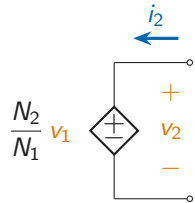
CCCS: $i_1 = -\frac{N_2}{N_1} i_2$ is $N_1 i_1 + N_2 i_2 = 0$.

VCVS: $v_2 = \frac{N_2}{N_1} v_1$.

A transformer is two dependent sources



CCCS, controlled by i_2



VCVS, controlled by v_1

CCCS: $i_1 = -\frac{N_2}{N_1} i_2$ is $N_1 i_1 + N_2 i_2 = 0$. VCVS: $v_2 = \frac{N_2}{N_1} v_1$.

Powers: $-\frac{N_2}{N_1} v_1 i_2$ and $+\frac{N_2}{N_1} v_1 i_2$ — one delivers exactly what the other absorbs.

Summary

- A **dependent source** (diamond) takes its value from a voltage or current in another branch: VCVS, CCVS, VCCS, CCCS. Unit of the constant: out over in.
- Its terminal equation names a foreign variable. Write KCL, KVL and terminal equations as usual, plus one equation for the controlling variable.
- It can **deliver power** — that is amplification — but it **cannot start anything**, and it is never switched off.
- **Ideal transformer**: $v_1/N_1 = v_2/N_2$, $N_1 i_1 + N_2 i_2 = 0$, signs set by the dots. Resistive, but a real one needs changing waveforms.
- It **absorbs no power at any instant** and rescales a load by $(N_1/N_2)^2$.

Next lecture

From circuit to graph

Labelling the nodes and branches of a circuit so that its $2b$ equations can be written systematically — the first step towards node and mesh analysis.

Questions?